

Git 101

徐延楷

What is Git?

- A version control system.
- Common Problem in team projects:

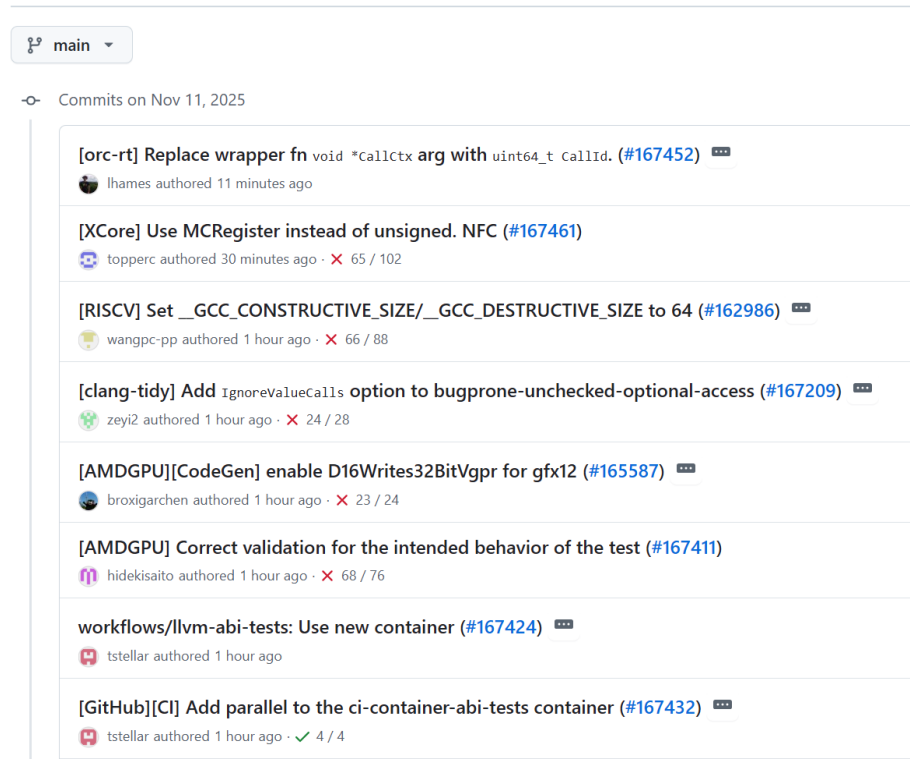


- Which zip is the latest version?
- My code is overwritten!
- You forget to merge my code in a file!
- The code doesn't work after changes, but I forgot what I did...
- Managing dozens of zips are painful.

What is Git?

- Git is a time machine for you project:

Commits



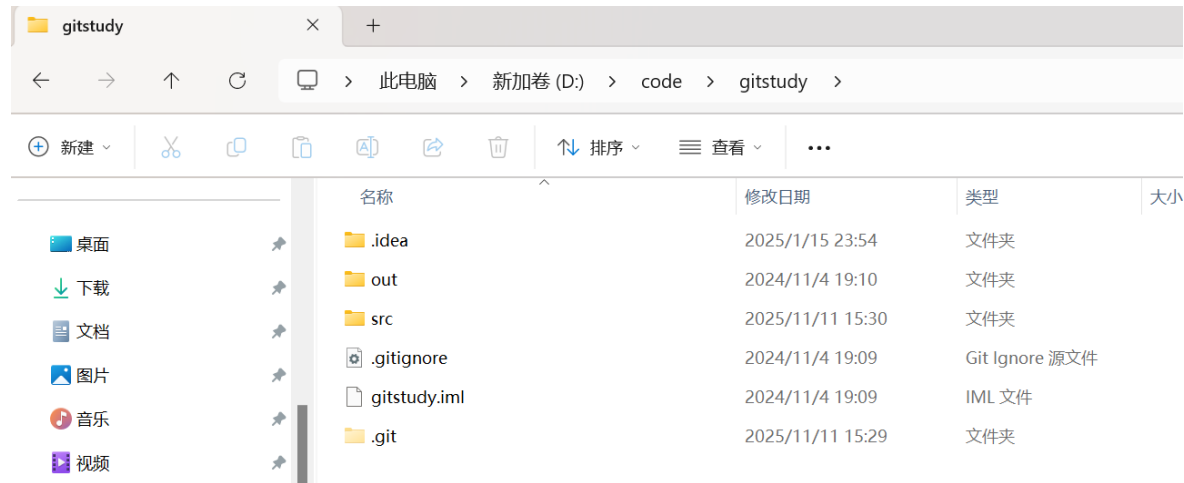
The screenshot shows a GitHub interface with the 'main' branch selected. Below the branch name, it says 'Commits on Nov 11, 2025'. A list of commits is displayed, each with a title, author, time, and a status icon (pass/fail).

- [orc-rt] Replace wrapper fn void *CallCtx arg with uint64_t CallId. (#167452)** by lhames, 11 minutes ago, status: pass.
- [XCore] Use MCRRegister instead of unsigned. NFC (#167461)** by topperc, 30 minutes ago, status: fail (65 / 102).
- [RISCV] Set __GCC_CONSTRUCTIVE_SIZE/_GCC_DESTRUCTIVE_SIZE to 64 (#162986)** by wangpc-pp, 1 hour ago, status: fail (66 / 88).
- [clang-tidy] Add IgnoreValueCalls option to bugprone-unchecked-optional-access (#167209)** by zeyi2, 1 hour ago, status: fail (24 / 28).
- [AMDGPU][CodeGen] enable D16Writes32BitVgpr for gfx12 (#165587)** by broxigarchen, 1 hour ago, status: fail (23 / 24).
- [AMDGPU] Correct validation for the intended behavior of the test (#167411)** by hidekisaito, 1 hour ago, status: fail (68 / 76).
- workflows/llvm-abi-tests: Use new container (#167424)** by tstellar, 1 hour ago, status: pass.
- [GitHub][CI] Add parallel to the ci-container-abi-tests container (#167432)** by tstellar, 1 hour ago, status: pass (4 / 4).

- Git helps:
- Keep a history of all changes
- Collaborate safely on the same files
- Recover from mistakes anytime
- Merge code from different person easily

Key concepts

- Repository (repo): Your project folder, and its history
- Commit: One version of your project. A “snapshot”.
- Github: A place to upload and share your repo to others, such as your teammates

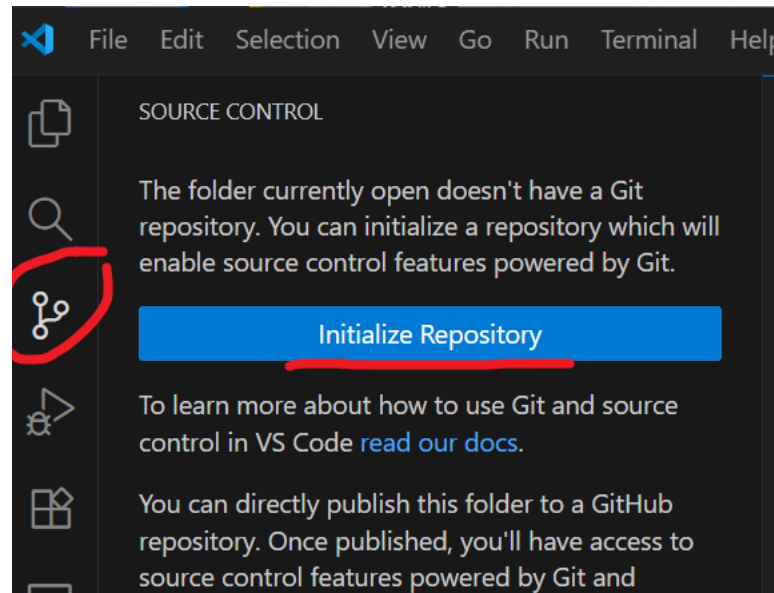


Git workflow

- You need to install git on your computer first
 - Goto <https://git-scm.com/install/>
- To use git, you can use vscode, Github Desktop, or command line
- I'll use vscode in this slides

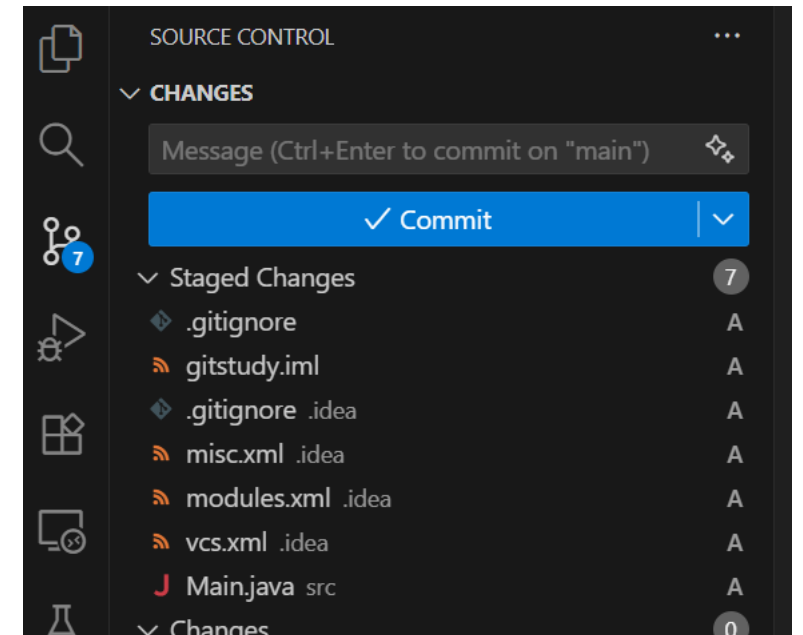
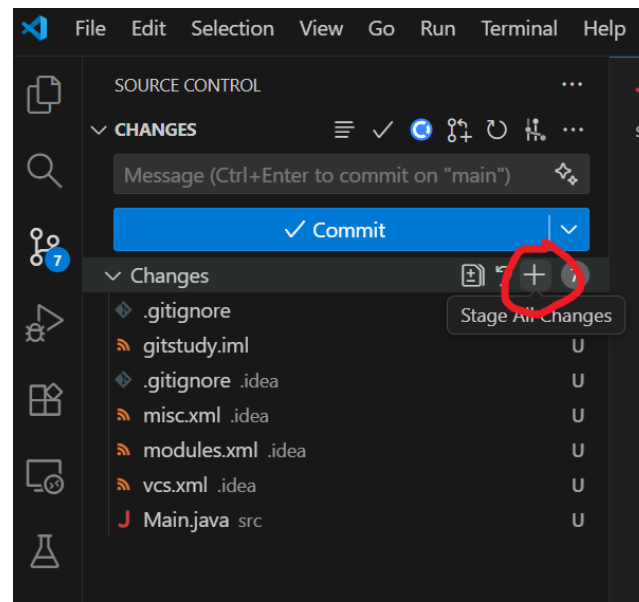
Git workflow

- First, you need to initialize the repo, to let git manage your folder
- Click git icon on the left, then click initialize Repository



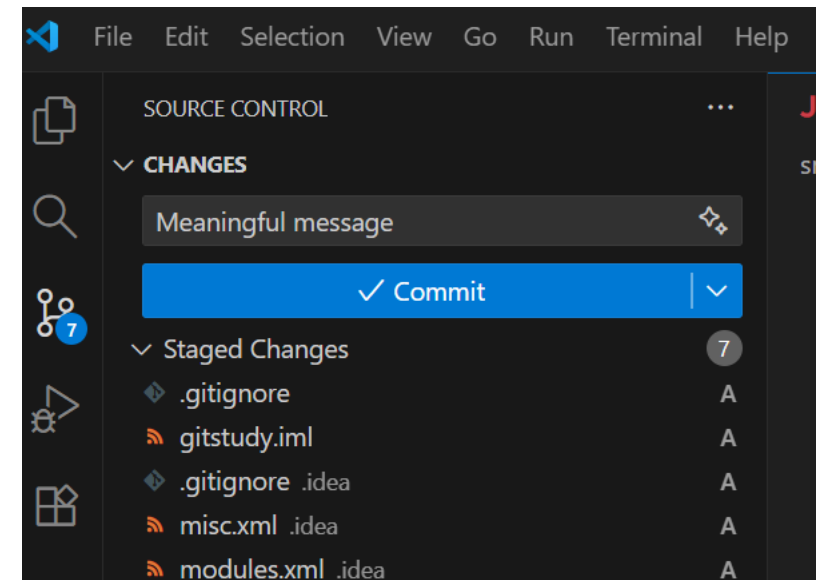
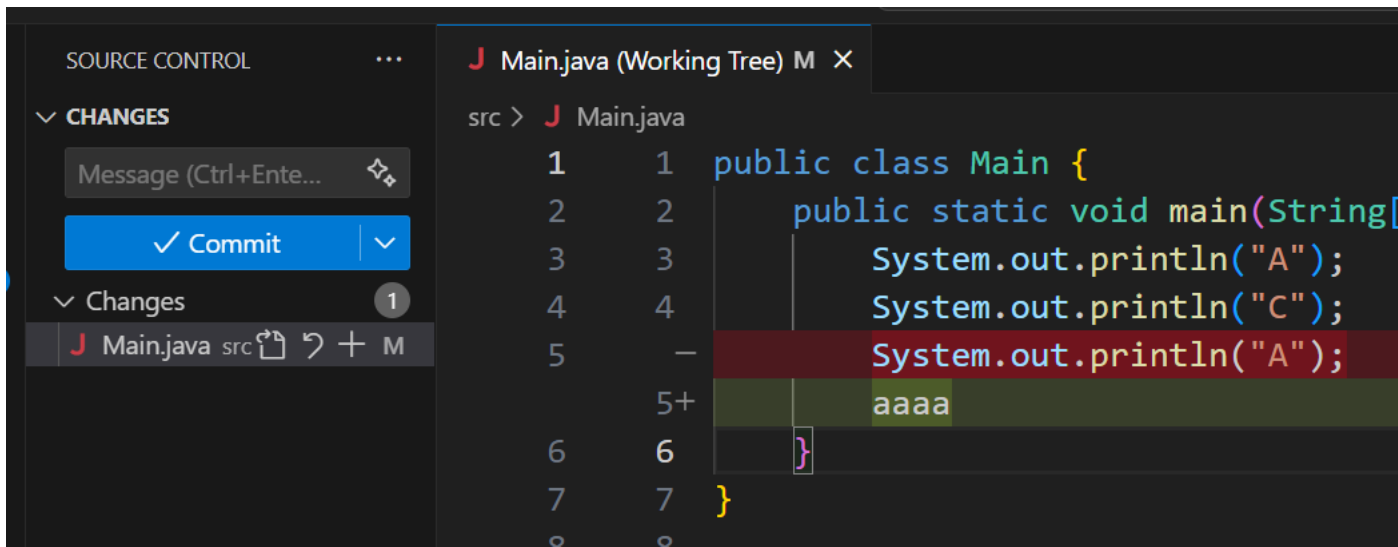
Git workflow

- Then, you can see your files in the repos.
- Move your mouse to “Changes”, then click the “+” mark.
- Files in “Changes” will go to “Staged Changes”, means they are ready to be tracked by git. (This is called “git add” or “stage”)



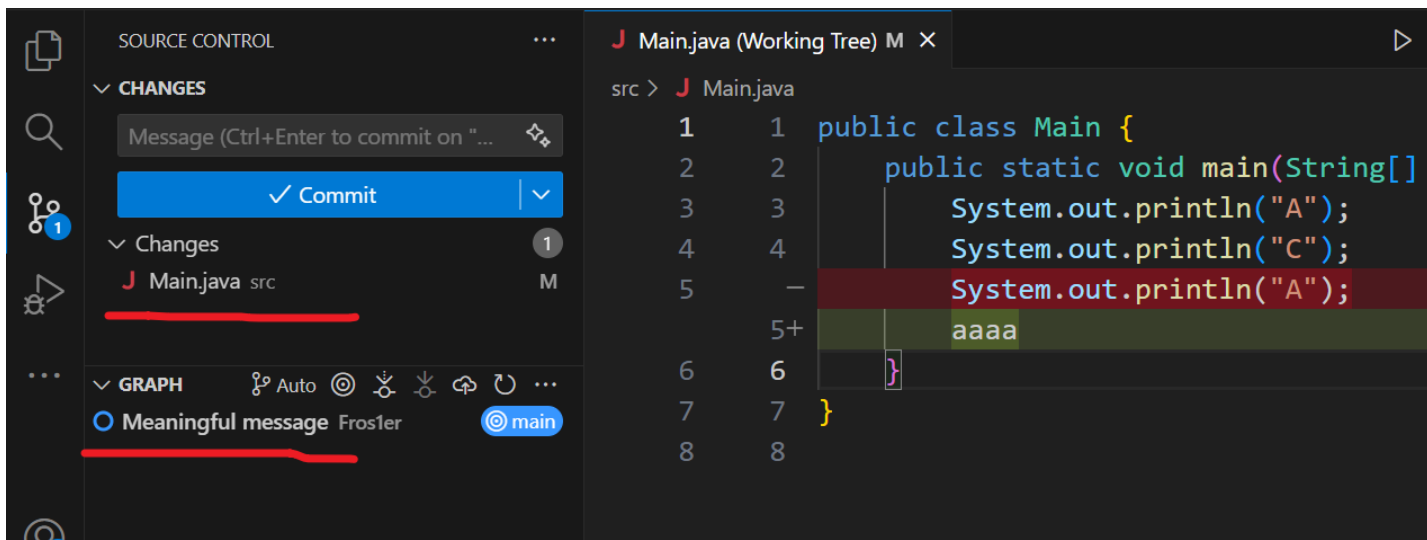
Git workflow

- If:
 - You want to track your further changes like below
 - You want to share your current code to your teammate
- Then you need a commit
- Type in some meaningful message, then click “Commit”



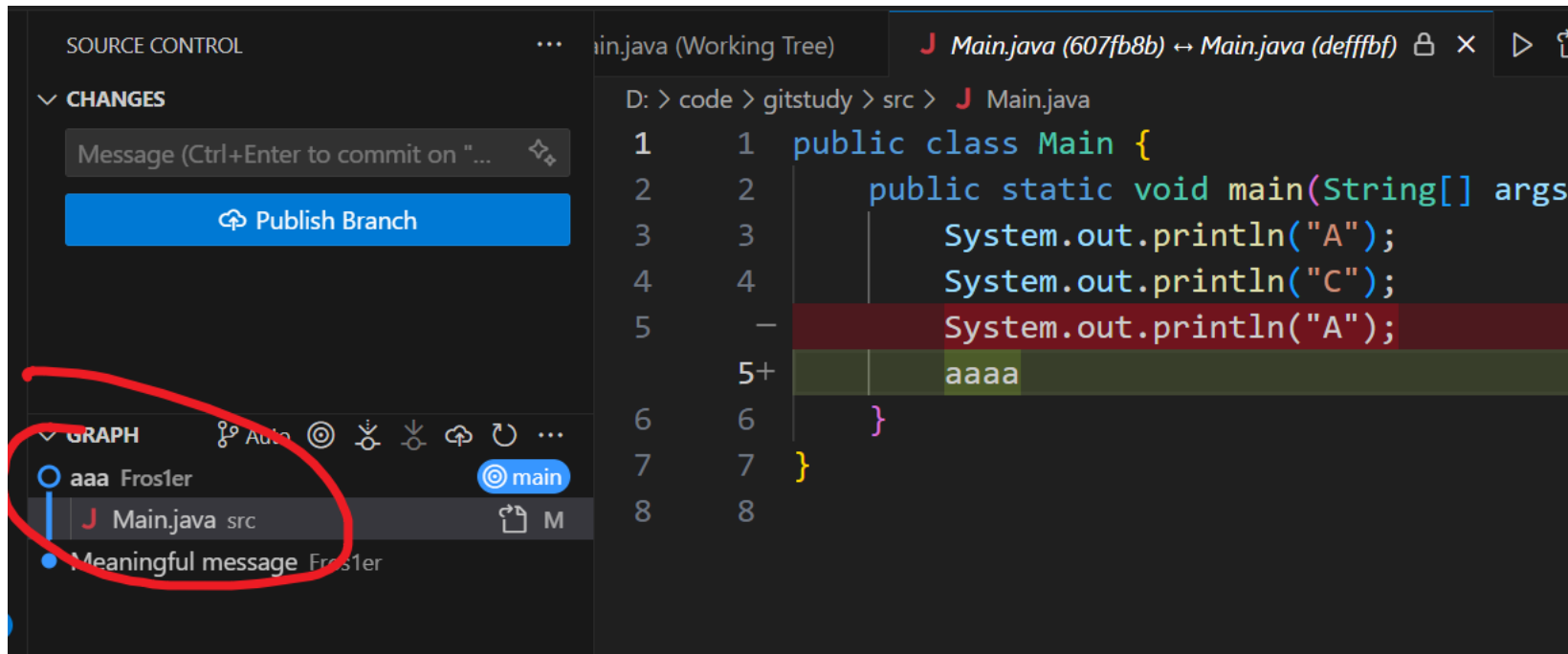
Git workflow

- You can see your commits in “Graph”
- Commit messages help you and your teammates track what a commit did
- If you made changes in your code, click file in “Changes” to see diff
- Commit frequently and remember to click “+” before commits!



Git workflow

- You can also click commits in “Graph”, to see code changes in your previous commits



Git workflow

- Git is far more powerful than just commits..
- You can search these if you are interested:
 - git branches
 - git checkout (go back to previous commit)
 - git reset (dangerous!)
 - Etc.
- This will help a lot for your collaborations

Upload repo to github

- For now, the repo only exists on your computer
- To collaborate with your teammates, you need to upload it to github
- First of all, you need a github account..
- Goto <https://github.com/#/> to sign up, and follow <https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent#/> to setup a key for git
- If you are the one who upload the repo, follow the steps:

Upload repo to github

- After signing up, you need to create a repo in github
- Although it's not the normal way to create a repo, you need to create the project repo by github classroom
- Goto <https://classroom.github.com/a/0vtGQe2t>
- Create or join a team, then go to the repo created (See appendix for detailed steps at end of slides)

CS207DigitalLogic-classroom-25Fall

Accept the group assignment —

Final project

Before you can accept this assignment, you must create or join a team. Be sure to select the correct team as you won't be able to change this later.

Create a new team:

Super awesome team name

+ Create team

You're ready to go —

test

You accepted the assignment, **Final project**.

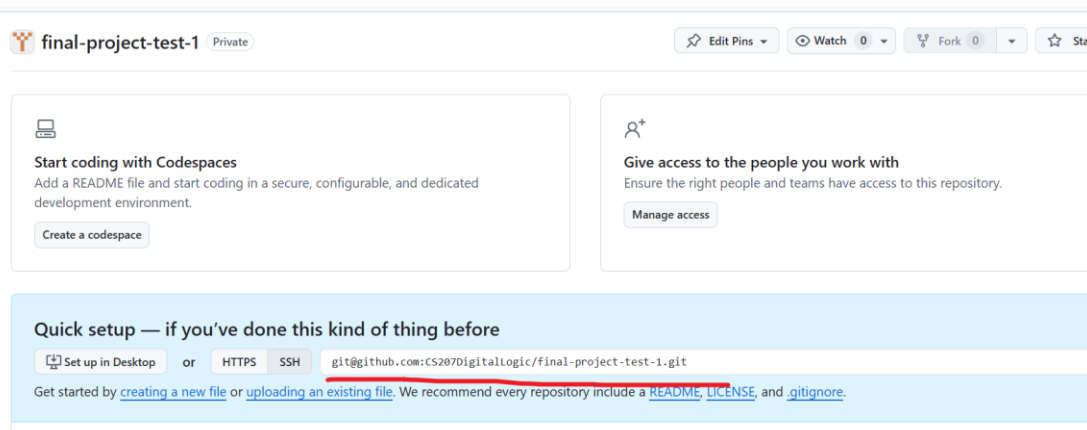
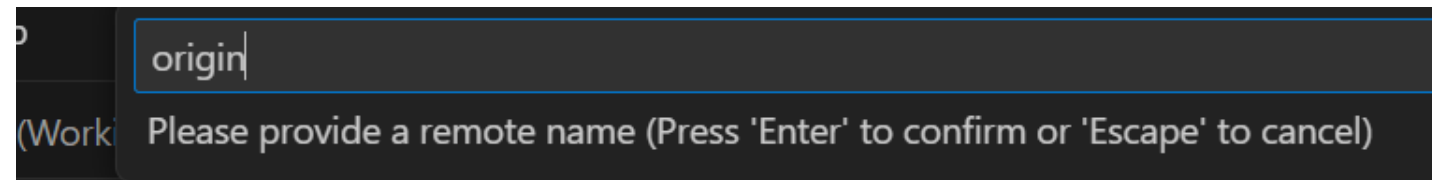
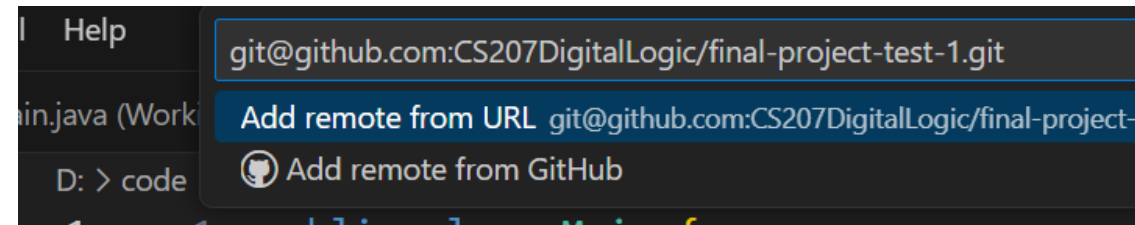
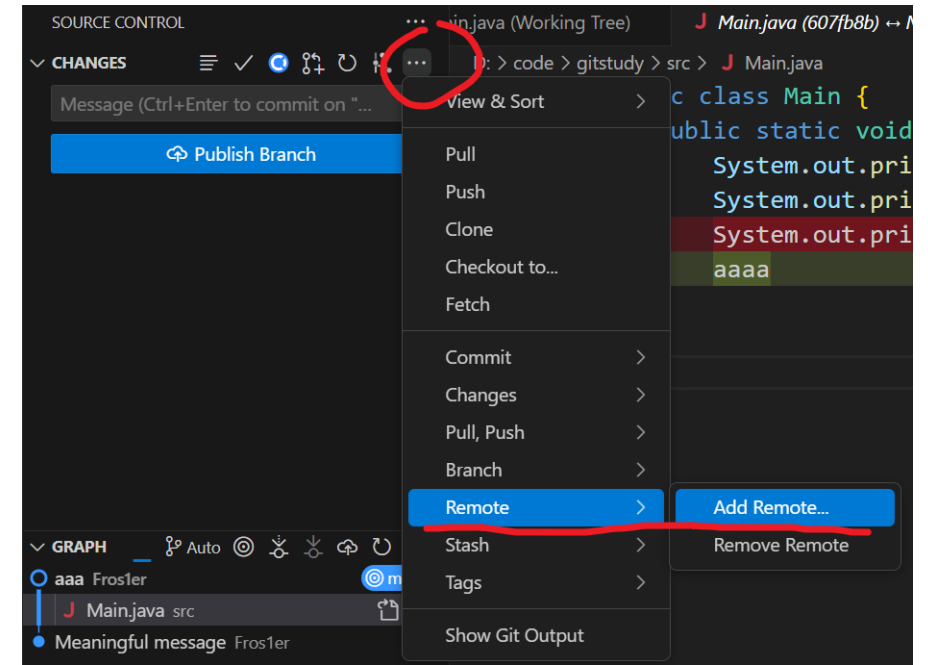
Your team's assignment repository has been created:

<https://github.com/CS207DigitalLogic/final-project-test-1>

We've configured the repository associated with this assignment.

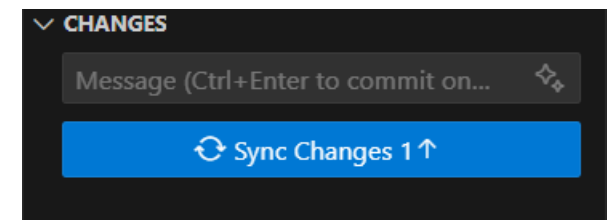
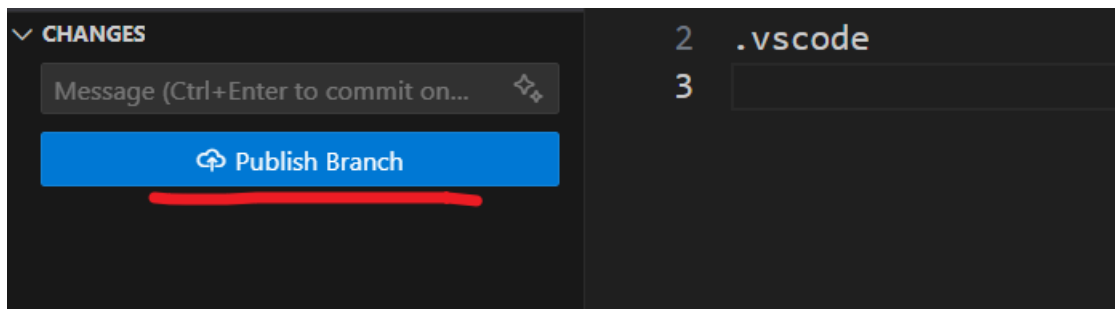
Upload repo to github

- Copy this link, and go back to vscode
- Click “Add Remote”
- Paste the url
- Type “origin”, enter
- Done



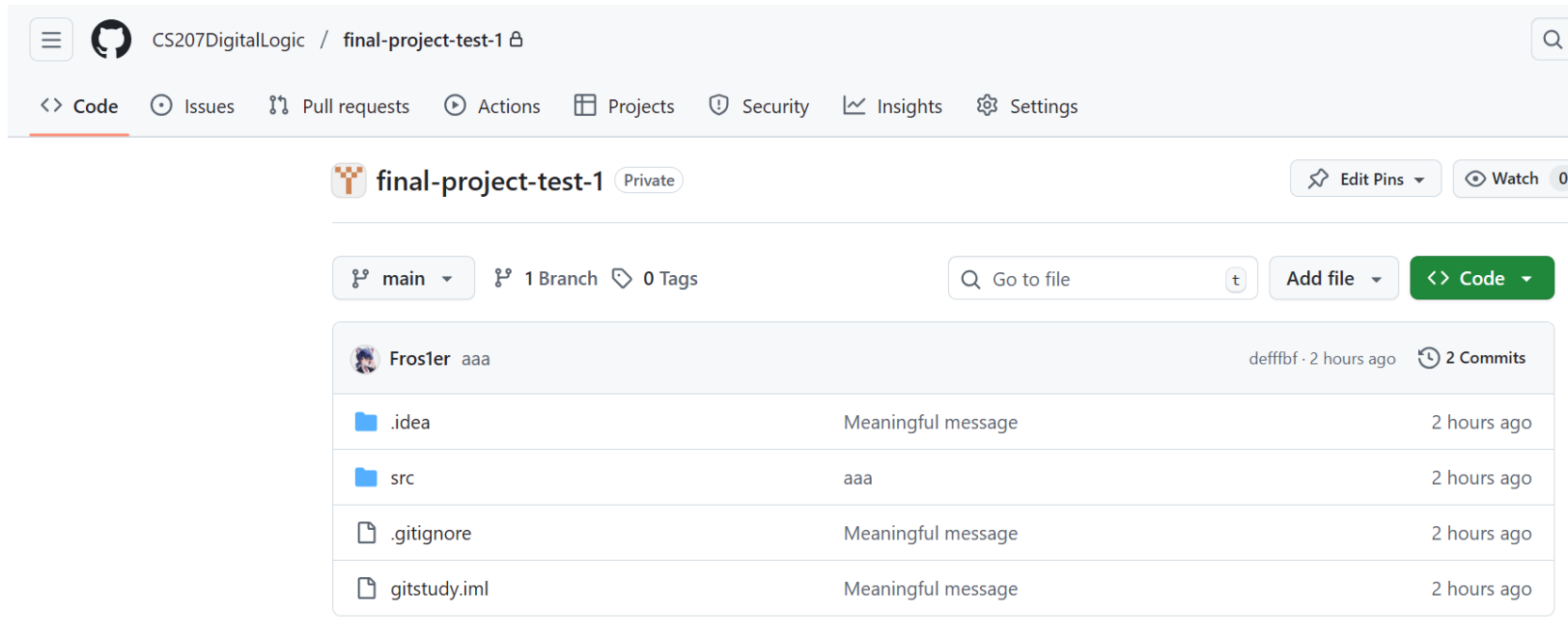
Push to github

- For now, we made a connection with local repo and github repo
- After each commit, we need to upload the commit to github repo
- This is called “push”.
- For the first push, click “Publish branch” after commit
- If you want to push again, click “Sync Changes”.



Push to github

- Then, your teammate can see your code in github!



The screenshot shows the GitHub interface for a repository named 'final-project-test-1' under the user 'CS207DigitalLogic'. The repository is private. The main branch is 'main'. The repository contains four files: '.idea', 'src', '.gitignore', and 'gitstudy.iml'. The commit history shows a single commit by 'Fros1er' with the message 'aaa' and a commit hash of 'deffbf', made 2 hours ago. The repository has 2 commits in total.

CS207DigitalLogic / final-project-test-1

<> Code Issues Pull requests Actions Projects Security Insights Settings

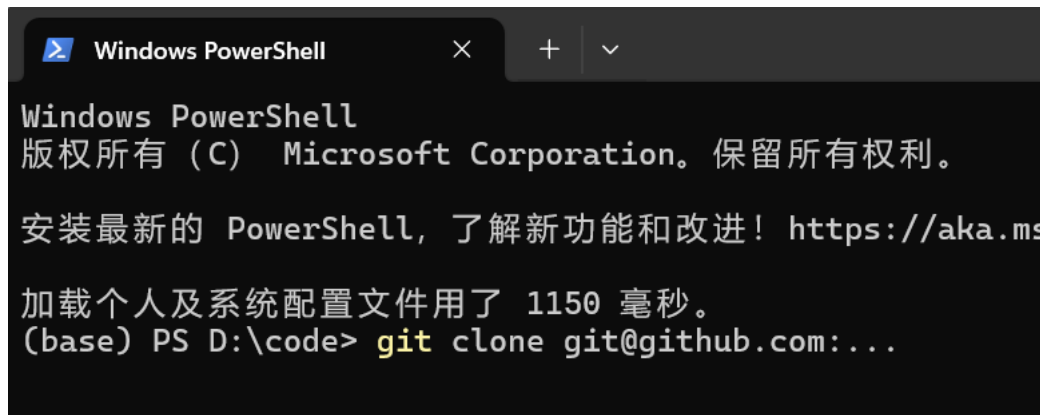
final-project-test-1 Private Edit Pins Watch 0

main 1 Branch 0 Tags Go to file Add file <> Code

File	Commit Message	Commit Hash	Time
.idea	Meaningful message	deffbf	2 hours ago
src	aaa	deffbf	2 hours ago
.gitignore	Meaningful message	deffbf	2 hours ago
gitstudy.iml	Meaningful message	deffbf	2 hours ago

Clone a repo from github

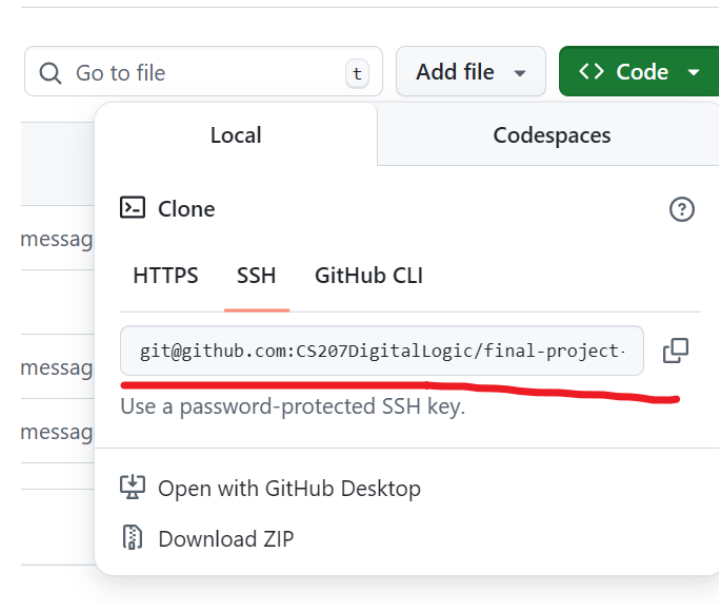
- To download the repo uploaded by your teammate, you need to clone the repo
- First, open the repo, copy the link of repo
- Open a terminal at where your repo want to be, type
`git clone <your link>`



```
Windows PowerShell
版权所有 (C) Microsoft Corporation。保留所有权利。

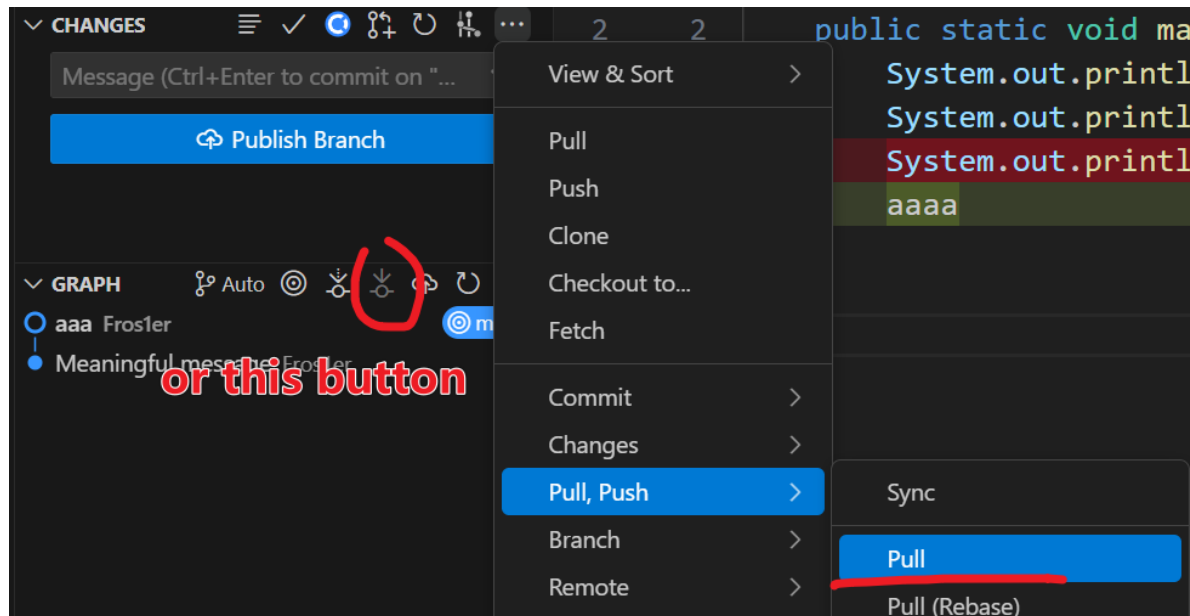
安装最新的 PowerShell, 了解新功能和改进! https://aka.ms/powershell

加载个人及系统配置文件用了 1150 毫秒。
(base) PS D:\code> git clone git@github.com:...
```



Pulls

- After you clone your repo, your teammate may make a commit and push again.
- This new commit **will not** automatically sync to your local repo!
- You need to pull them to your computer.



Merge conflict

- Sometimes:
 - A committed “bbbb” with “commit 1” and pushed
 - B committed “cccc” with “commit 2” without pull “commit 1” first
 - What will happen if B pull after “commit 2”?

 Fros1er commit 1

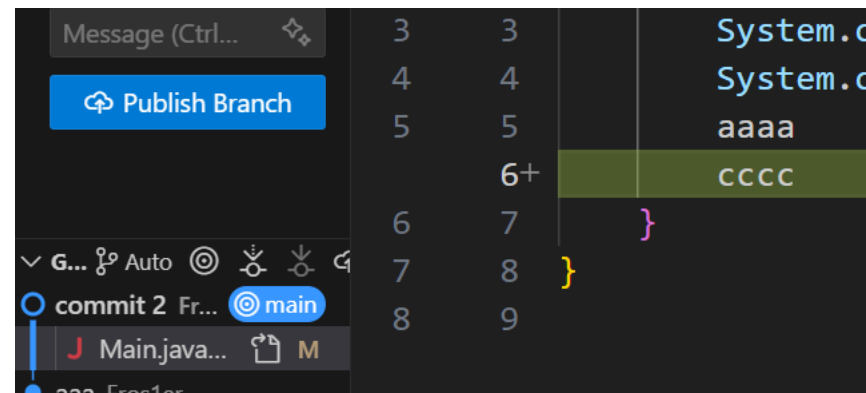
Code

Blame

8 lines (8 loc) · 169 Bytes

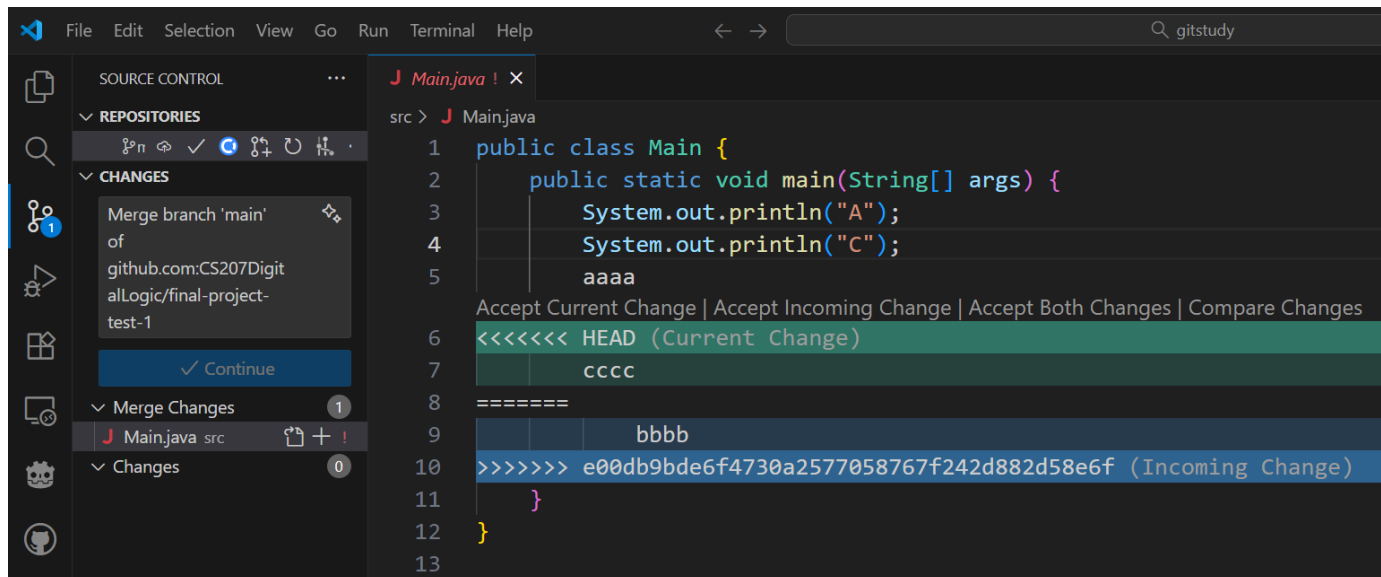


```
1  public class Main {
2      public static void main(String[] args) {
3          System.out.println("A");
4          System.out.println("C");
5          aaaa
6          bbbb
7      }
8  }
```



Merge conflict

- When pull, git will merge the files automatically if it's ok
 - E.g. when two commits changes different files
- However, two commits are editing the same place in this example
- Need to manually solve a “Merge conflict”, as git doesn't know which code to accept



```
File Edit Selection View Go Run Terminal Help
src > J Main.java
1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("A");
4         System.out.println("C");
5         aaaa
6 <<<<<< HEAD (Current Change)
7         cccc
8 =====
9         bbbb
10 >>>>>> e00db9bde6f4730a2577058767f242d882d58e6f (Incoming Change)
11     }
12 }
13
```

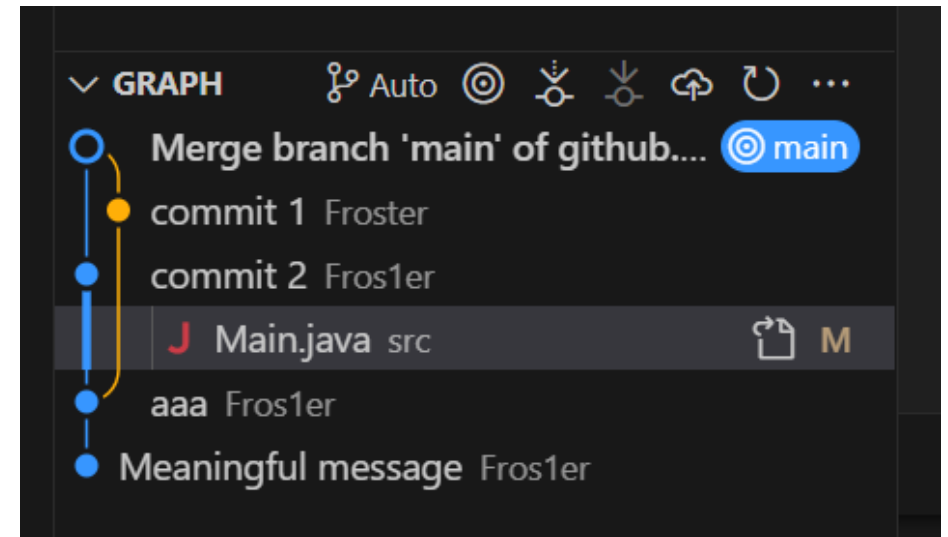
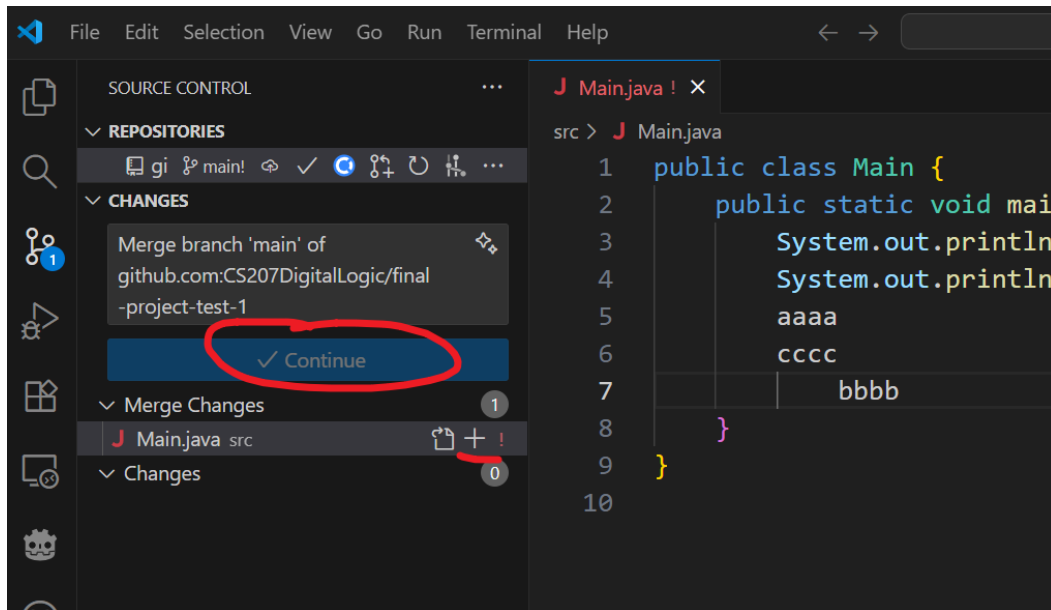
Solve merge conflict

- Solving merge conflict means editing the file, and leave what you want.
- You can remove the lines with “<<<<” and codes directly, or use buttons for quicker actions.

```
src > J Main.java
1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("A");
4         System.out.println("C");
5         aaaa
6 <<<<<< HEAD (Current Change)
7     cccc
8 =====
9         bbbb
10 >>>>>> e00db9bde6f4730a2577058767f242d882d58e6f (Incoming Change)
11     }
12 }
```

Solve merge conflict

- After editing the file, stage it and click “continue” to make a new commit.
- You can see the diverged commits and the merge clearly in graph.
- Don't forget to push it!

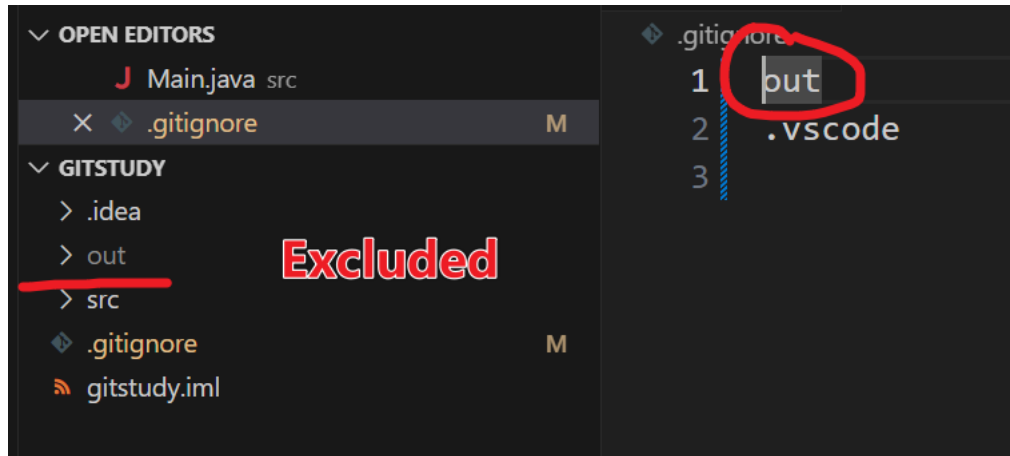


Best practices for using git

- Write clear commit messages. E.g. “action: summary”
- Make commit when:
 - You want some history to help with your development (Don’t push here!)
 - The code is relatively okay and runnable (Push as soon as possible)
 - Sometimes the code don’t need to be perfect. You need to frequently merge with your group.
- Always pull before you start to develop a new part
- Communicate before start. Sometimes it’s better to wait your teammate finish and push.

.gitignore

- Some files and folders, like “.vscode” or “xx.runs” (synth and implementation result of vivado), shouldn’t be managed by git
- They change rapidly and you don’t need their history!
- Add a .gitignore file at root of project to exclude these folders.

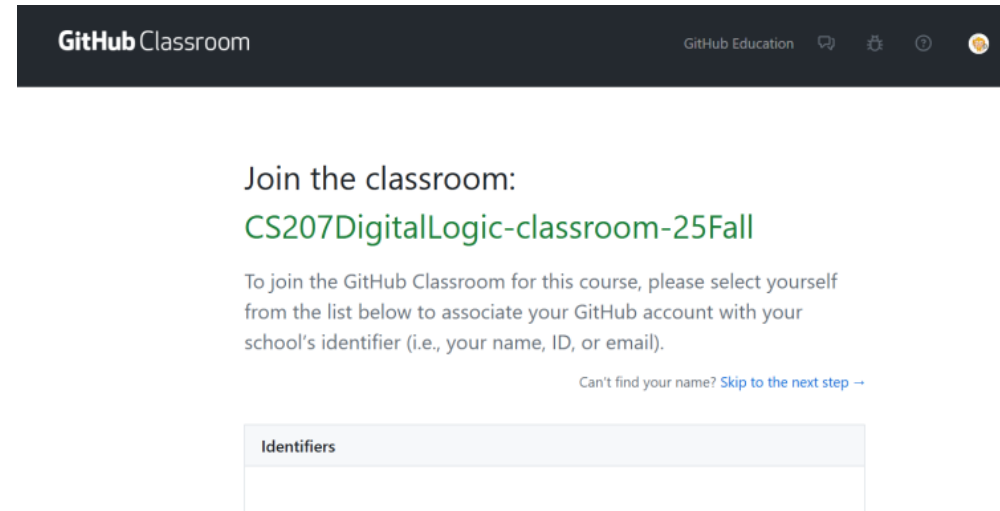


.gitignore examples for vivado projects:

- https://adaptivesupport.amd.com/s/article/61232?language=en_US#/ (by amd)
- <https://gist.github.com/iDoka/6147ea6cf320f09d7ca39477b090486a#/>

Appendix for github classroom

- To create a repo, goto <https://classroom.github.com/a/0vtGQe2t>
- Click “Skip to the next step”



Appendix for github classroom

- Join or create a team
- For who creates the team, try find a good team name so your teammate can find you
- After then your repo should be ready
- All the teammates will share the same repo

